

Execution-Aware Ichimoku + EMA Backtest on Indonesian Commodity Stocks

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1. Overview

This project tests a long-only daily trend-following strategy on five Indonesian commodity and energy stocks: PTBA.JK, ANTM.JK, BRPT.JK, MDKA.JK, and PGAS.JK. The strategy combines Ichimoku Cloud trend confirmation, EMA 13/26 filtering, and commission-adjusted backtesting to see whether a simple rules-based system can survive realistic execution costs in the Indonesian market.

The project was built as a quant portfolio piece to show end-to-end research work: data collection, indicator engineering, signal generation, backtesting, performance analysis, and interpretation. The first version of the results was mixed, which is still useful because it highlights that the strategy is selective rather than universally strong.

2. Research Objective

The core question is simple: can a disciplined technical strategy generate usable risk-adjusted returns on liquid Indonesian commodity stocks after transaction costs are included?

The project is not trying to prove that Ichimoku or EMA alone is a magic edge. Instead, it studies whether the combination of trend confirmation, signal filtering, and execution-aware backtesting can identify stocks where a long-only approach is more effective than passive holding.

3. Market Universe

The research universe contains five Indonesian commodity and energy names:

- PTBA.JK
- ANTM.JK
- BRPT.JK
- MDKA.JK
- PGAS.JK

These stocks were chosen because they are relatively liquid, locally relevant, and naturally tied to a sector theme that is easy to explain in a portfolio setting. Using a consistent basket also makes it easier to compare performance across names rather than studying isolated one-off trades.

4. Data and Assumption

The project uses daily OHLCV data downloaded from Yahoo Finance. The backtest starts from 2018 and applies a simple commission assumption of 0.25% per trade. The strategy is long only, and when the signal turns off, the position goes to cash rather than short.

That choice is intentional. The Indonesian market context makes a long-only setup more realistic for a retail-style project, and it also keeps the implementation cleaner and easier to explain. Buy-and-hold is used as the benchmark for every stock.

5. Strategy Design

Entry Rules

A long position is opened only when all of the following are true:

- Close price is above the Ichimoku cloud.
- Tenkan-sen is above Kijun-sen.
- EMA 13 is above EMA 26.
- Optional: volume is above its 20-day moving average.

Exit Rules

The position is closed when any of the following happens:

- Close price falls into or below the cloud.
- EMA 13 falls below EMA 26.
- Tenkan-sen falls below Kijun-sen.

This structure was designed to do two things at once. First, it tries to capture trend persistence. Second, it avoids staying in a trade once the trend weakens. The strategy is deliberately simple so the logic can be tested and explained without overfitting the whole project around a large parameter search. The schematic diagram of the strategy design shown in figure 1.

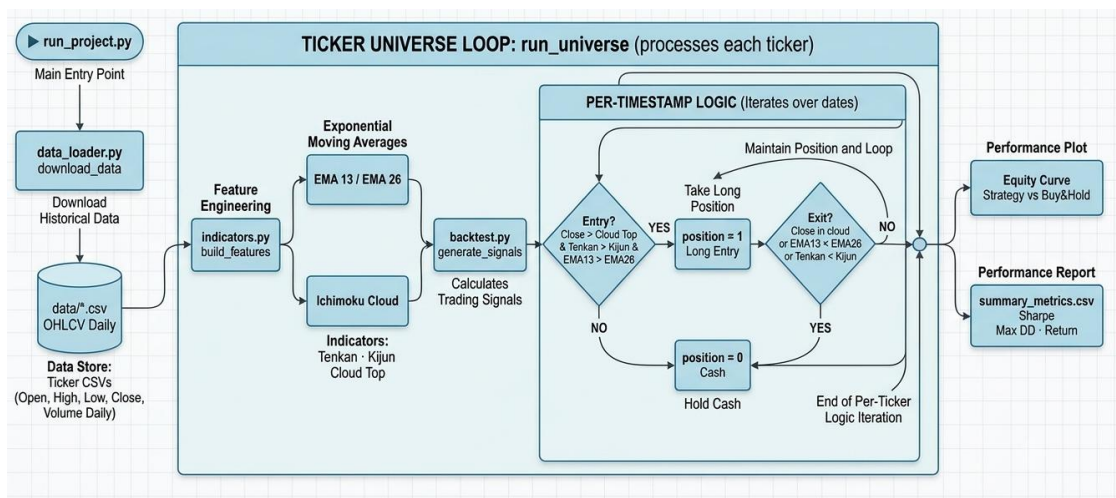


Figure 1 Quantitative Trading schematic workflow using Ichimoku Cloud and EMA13/EMA26 Crossing.

6. Methodology

The workflow was built in Python and split into a few clean steps. First, the raw data is downloaded and saved. Then the indicators are calculated. After that, entry and exit signals are generated, and the backtest is run as a long-only simulation with commissions included.

Performance is measured using total return, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, number of trades, and days in market. The strategy curve is compared with buy-and-hold so the result is not judged in isolation.

7. Results Summary and Key Findings

The strategy produced mixed results across the five stocks. It performed best on BRPT.JK, where the strategy clearly outperformed buy-and-hold on total return and showed the strongest risk-adjusted result in the basket. ANTM.JK was positive on the strategy side but still lagged its passive benchmark.

PGAS.JK and PTBA.JK were only mildly positive and did not clearly beat buy-and-hold. MDKA.JK was the weakest result, ending with a negative total return and a negative Sharpe ratio. In other words, the strategy is not universal. It seems to behave very differently depending on the stock.

- BRPT.JK was the strongest performer in the basket.
- ANTM.JK had positive strategy return but weak risk-adjusted quality.
- PTBA.JK and PGAS.JK were marginal and did not strongly beat passive holding.

- MDKA.JK was the weakest result in the group.
- Drawdowns were still large in several cases, which means the strategy needs refinement.

Additional information for this results shown in `/output/charts/[stocks name].png`

8. Interpretations

The most important lesson is that the strategy is selective, not universal. That is a real finding, not a failure. A simple trend-following setup can work well on some Indonesian commodity stocks, but not all of them, and not in every market regime.

The results suggest that stock-specific behavior matters a lot. BRPT.JK may have had stronger trend persistence or cleaner directional phases, while the other names may have been noisier, more mean-reverting, or simply less friendly to the chosen rules. The large drawdowns also show that signal quality alone is not enough; the exit design and risk controls matter just as much.

9. Limitations and Future Recommendations

This project is intentionally simple, so it has several limitations:

- It does not use short selling.
- It does not use leverage.
- It does not model market impact in depth.
- It does not optimize parameters automatically.
- It does not guarantee generalization across different regimes.

Another limitation is that the first version uses a fixed rule set. That is fine for a portfolio project because the main purpose is to demonstrate research discipline, but the next version could explore whether adding a regime filter or a volume confirmation improves robustness.

Possible improvements include:

- Adding a volume confirmation filter.
- Testing a weekly trend filter on top of the daily signal.
- Adding ATR-based stop loss or trailing stop logic.
- Testing more Indonesian stocks in the same sector theme.
- Studying whether BRPT.JK's stronger behavior is linked to volatility regime changes or trend persistence.

10. Conclusion

This project shows a practical first step into systematic trading research on Indonesian stocks. The current results are mixed, but that is exactly what makes the work interesting and credible. It demonstrates technical analysis implementation, backtesting discipline, and the ability to evaluate a strategy honestly under realistic market constraints.